QP CODE: 25900084

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Multi-DisciplinaryCourses (MDC) - MG2MDCFTQ100 - FOOD ADDITIVES

(2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)
Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Short Answer Questions Answer any 5 questions
Each question carries 1 marks

1. **Which method is used to extract colorants from citrus fruits? [K] / [CO5]**
   1. Solvent Extraction: The citrus peels or juice are treated with organic solvents like ethanol, methanol, acetone, or hexane to dissolve the colorants.
   2. Supercritical Fluid Extraction (SFE): Uses supercritical CO_2 (often with ethanol as a co-solvent) to extract colorants under high pressure and temperature.
   3. Ultrasound-Assisted Extraction (UAE): Ultrasonic waves break cell walls, enhancing solvent penetration and release of pigments.
   4. Microwave-Assisted Extraction (MAE): Microwaves heat the plant matrix, improving the efficiency of solvent-based extraction.

2. Name 2 sequestrants. **[K] / [CO4]**
Two common sequestrants used in food processing are: 1. Citric acid 2. EDTA (Ethylenediaminetetraacetic acid)

3. Give two examples of polyols. **[K] / [CO5]**
Two common examples of **polyols** (also known as sugar alcohols) are:

1. **Sorbitol**
2. **Xylitol**

4. **What is one carrier material used in the microencapsulation process? [U] / [CO5]**

   One commonly used **carrier material** in the microencapsulation process is **maltodextrin**
5. **Name one example of a natural hydrocolloid derived from seaweed. [K] / [CO5]**

One example of a natural hydrocolloid derived from seaweed is **agar**.It's commonly used as a **gelling agent** in foods like jellies, desserts, and even in microbiological media.

6. **What are the uses of corn sweeteners in food production? [K] / [CO5]**

Corn sweeteners have several important uses in food production, including:

1. Sweetening 2. Enhancing texture 3. Preservation 4. Fermentation 5. Color and flavor development –

7. **Name one common antifoaming agent used in food processing. [K] / [CO3]**

One common antifoaming agent used in food processing is dimethylpolysiloxane.[1x5 =5]

Part B

Short Essay Questions Answer any 5 questions

Each question carries 4 marks

8. **Classify food additives based on their function and provide examples for each category. [K] / [CO1]** 1. Preservatives: Prevent spoilage caused by microorganisms and extend shelf life.
Examples: Sodium benzoate, Potassium sorbate

2. Antioxidants: Prevent oxidation, which can cause rancidity or color changes.Examples: Ascorbic acid (Vitamin C), Butylated hydroxyanisole (BHA)

3. Colorants (Food Colors): Enhance or add color to food.
Examples: Tartrazine (synthetic)Beet juice (natural)
4. Flavor Enhancers: Improve or intensify flavor without adding their own.
Examples: Monosodium glutamate (MSG), Disodium inosinate
5. Sweeteners: Provide sweetness, often with fewer calories than sugar.
Examples: Aspartame (artificial), Stevia (natural)
6. Emulsifiers: Help mix ingredients that normally don't blend (like oil and water).
Examples: Lecithin, Mono- and diglycerides
7. Stabilizers, Thickeners, and Gelling Agents: Improve texture and consistency.
Examples: Guar gum Carrageenan
8. Acidity Regulators: Control the pH of food to enhance flavor or preservation.
Examples: Citric acid, Lactic acid
9. Anti-caking Agents: Prevent clumping in powdered or granulated products.
Examples: Silicon dioxide, Calcium silicate
10. Antifoaming Agents: Reduce or prevent foaming during food processing.
Examples: Dimethylpolysiloxane

9. How do preservatives help prevent oxidation in food? [K] / [CO3]

Preservatives help prevent oxidation in food by slowing down or stopping the chemical reactions between oxygen and food components like fats, oils, and vitamins. These reactions can cause: Rancidity (bad smells and flavors), Color changes (like browning or fading), Nutrient loss (especially of vitamins A, C, and E. Some preservatives act as antioxidants themselves (like ascorbic acid or BHA) by:Binding oxygen before it can react with food, Neutralizing free radicals that cause oxidative damage

10. What do you understand by antimicrobial agents? Provide examples. [K] / [CO2]

Antimicrobial agents are substances added to food to prevent the growth of harmful microorganisms such as bacteria, yeasts, and molds. Their main role is to extend shelf life and enhance food safety by reducing the risk of spoilage and foodborne illness.

 Examples of Antimicrobial Agents:

1. Sodium benzoate
    - Used in: Soft drinks, fruit juices, pickles
    - Effective against: Yeasts and molds
2. Potassium sorbate
    - Used in: Baked goods, cheeses, syrups
    - Effective against: Molds, yeasts, and some bacteria
3. Nisin (a natural peptide)
    - Used in: Processed cheese, canned foods, meat products
    - Effective against: Gram-positive bacteria
4. Natamycin
    - Used in: Cheese and baked goods
    - Effective against: Fungi and molds

11. What are the benefits of using honey in recipes instead of sugar? [K] / [CO5]

Using honey instead of sugar in recipes can offer several benefits, both in terms of flavor and nutrition. Here are some of the key advantages:1. Natural Sweetener.2. Higher Nutrient Content 3. Moisture Retention4. Lower Glycemic Index5. Distinct Flavor Profile6. Antimicrobial and Health Benefits7. Fewer Calories per Teaspoon

12. How do buffering salts help maintain flavor and prevent extreme pH changes in food? [K] / [CO5]

Buffering salts play an important role in maintaining the **pH stability** of foods, which helps to **preserve flavor** and prevent extreme pH changes that could affect both the taste and the quality of the product

13. What are the uses of flavorings and sweeteners in the food industry. [K] / [CO5]

In the food industry, flavorings and sweeteners are used extensively to enhance the sensory experience of food and beverages. Here's a breakdown of their uses:

Uses of Flavorings:

1. Enhance or Modify Taste
2. Mask Off-Flavors
3. Increase Variety and Appeal
4. Create Signature Tastes

Uses of Sweeteners:

1. Provide Sweetness with Fewer Calories
2. Control Blood Sugar Levels
3. Enhance Product Texture
4. Extend Shelf Life
5. Mask Unpleasant Aftertaste
6. Improve Fermentation Processes

14. How does citric acid help preserve the color of fruits and vegetables? Provide examples of its application in food.[K] / [CO5]

Citric acid plays a crucial role in preserving the color of fruits and vegetables by acting as a pH regulator and antioxidant.How Citric Acid Preserves Color:

1. Prevents Browning (Enzymatic Browning)
2. Antioxidant Properties
3. Prevents Loss of Vitamin C
4. Enhances Fruit and Vegetable Brightness

Applications of Citric Acid in Food:

1. In Canned Vegetables and Fruits
2. In Juices and Beverages
3. In Frozen Foods
4. In Jams and Jellies

[4x5 = 20]

Part C

Long Essay Questions Answer any 1 questions Each question carries 10 marks

15. Discuss the role of acidulants and anticaking agents in food production? [U] / [CO5]

. Acidulants are food additives specifically used to lower the pH of food products. They're organic or inorganic acids that serve multiple purposes in food processing beyond simply making foods taste sour. Roles in Food Production:1. Flavor Enhancement 2. Preservation 3. pH Control. 4. Synergistic Effects 5. Texture and Gel Formation

Common Examples: Citric acid, acetic acid, lactic acid, malic acid, phosphoric acid.

Anticaking Agents: Anticaking agents are substances added to powdered or granulated foods to prevent clumping and maintain free-flowing properties.

Roles in Food Production: Flow Improvement, Product Consistency and Shelf Life Extension

Common Examples: Silicon dioxide, calcium silicate, magnesium carbonate, tricalcium phosphate.

16. How do leavening agents contribute to the baking process? Discuss their impact on the texture and volume of baked goods. [K] / [CO5]

Leavening agents are essential in the baking process because they help baked goods rise, resulting in a light, airy texture and increased volume. Without leavening agents, most baked items like bread, cakes, and muffins would be dense and flat.

Impact on Texture and Volume:

- Volume: Gas expansion causes the dough or batter to rise, increasing the final size.
- Texture:

Creates a light, spongy or airy crumb.

Prevents dense or tough products.

Affects crumb structure (fine vs. coarse depending on the agent used and mixing technique).

17. **What is the mechanism by which antioxidants prevent the oxidation of food components? Discuss how they contribute to maintaining food quality during storage. [K] / [CO5]**Antioxidants work by slowing or preventing the oxidation of food components, particularly fats, oils, and certain vitamins, which are prone to degradation when exposed to oxygen, light, heat, or metal ions. Their main role is to protect food quality, extend shelf life, and preserve nutritional value during storage.Antioxidants play a crucial role in maintaining food quality during storage by preventing or slowing down oxidative reactions that can deteriorate the taste, smell, color, texture, and nutritional value of food.

1. Prevent Rancidity in Fats and OilsAntioxidants such as BHA, BHT, or tocopherols (Vitamin E) block this reaction by neutralizing the free radicals that trigger fat oxidation.
2. Preserve Color and Appearance
3. Maintain Nutritional Value
4. Extend Shelf Life
5. Stabilize Flavors and Aromas

[10x1 = 10]